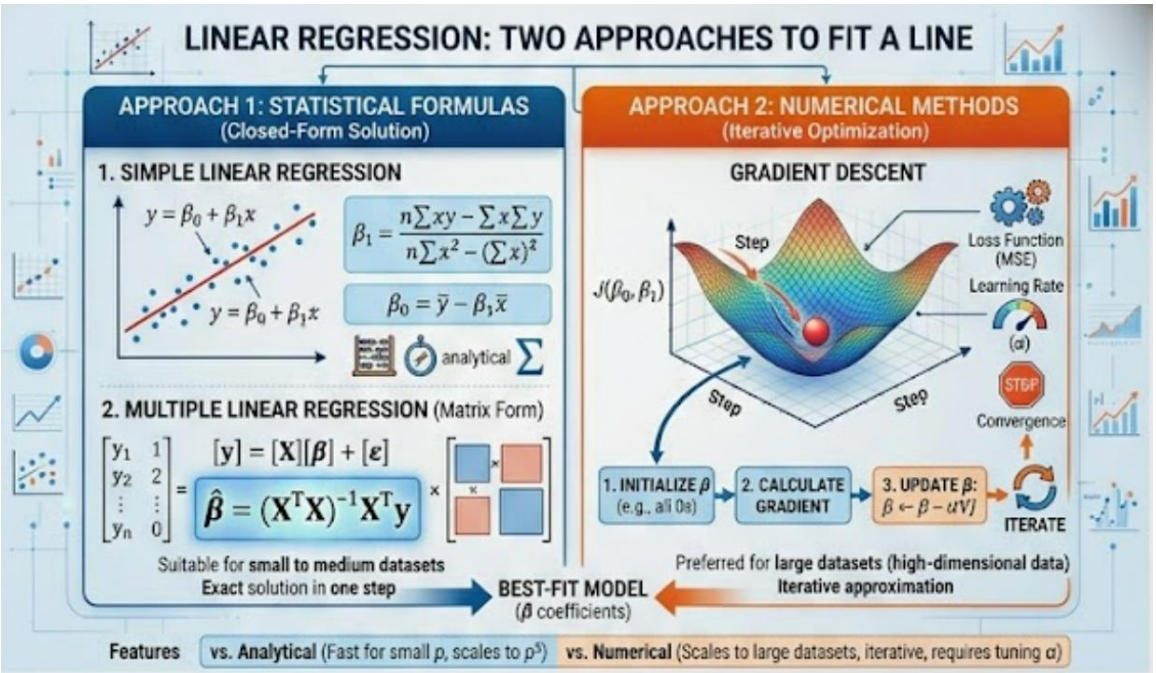


Linear Regression Using Numerical Methods and Statistical Formulas

Linear Regression Using Numerical Methods
Statistical Formula-Based Implementation (No ML Libraries)

Subject: Numerical Methods
Submitted by: Yash Lund, Tushar Sharma, Surya Makwana
Dataset: USA Housing Prices
Platform: Python (From Scratch Implementation)

GitHub link: https://github.com/yashlund05/House-price-prediction--linear-regression-.git

1. Introduction

Linear Regression is a fundamental statistical technique used to model the relationship between an independent variable (X) and a dependent variable (Y). In this project, linear regression is implemented purely using numerical methods and statistical formulas, without using any machine learning libraries.

The objective is to compute all regression parameters manually using summations, deviations, correlation formulas, and error analysis, and to visualize the results using plots.

The dataset used is the USA Housing Prices dataset from Kaggle.

2. Dataset Description

Dataset: Housing.csv (USA Housing Prices)

Features:

- Avg. Area Income
- Avg. Area House Age
- Avg. Area Number of Rooms
- Avg. Area Number of Bedrooms
- Area Population
- Price (target variable)
- Address (dropped as non-numeric)

For simple linear regression:

X = Avg. Area Income

Y = Price

Total samples after cleaning: 4977

3. Data Cleaning (Numerical Method)

Z-score method was used for outlier removal:

$Z = (X - \mu) / \sigma$

Outliers where $|Z| > 3$ were removed.

Original samples: 5000

Outliers removed: 23

Final samples: 4977

4. Statistical Calculations

Mean:

$\mu = (1/n) * \Sigma Xi$

Median:

Middle value after sorting.

Mode:

Estimated by binning continuous data.

Variance and Standard Deviation:

$\sigma^2 = (1/n) * \Sigma(Xi - \mu)^2$

$\sigma = \sqrt{\sigma^2}$

5. Correlation and Regression (From Formulas)

Pearson Correlation Coefficient:

$r = \Sigma[(Xi - \bar{X})(Yi - \bar{Y})] / \sqrt{[\Sigma(Xi - \bar{X})^2 * \Sigma(Yi - \bar{Y})^2]}$

$r = 0.631255$

Regression Coefficients:

Slope:

$m = \Sigma[(Xi - \bar{X})(Yi - \bar{Y})] / \Sigma[(Xi - \bar{X})^2]$

Intercept:

$c = \bar{Y} - m\bar{X}$

Regression Equation:

$\hat{Y} = 20.9507X - 204809.23$

6. Prediction and Residuals

Residual:

$ei = Yi - \hat{Y}i$

Absolute Error and Squared Error were computed for all samples.

7. Error Metrics

Mean Absolute Error (MAE):

$MAE = (1/n) * \Sigma|ei|$

Mean Squared Error (MSE):

$MSE = (1/n) * \Sigma(ei)^2$

Root Mean Squared Error (RMSE):

$RMSE = \sqrt{MSE}$

Coefficient of Determination:

$R^2 = 1 - (SSres / SStot)$

Metric Values:

MAE = 216,373.12

MSE = 72,796,455,341.79

RMSE = 269,808.18

R^2 = 0.398483 (39.85% variance explained)

8. Graphical Analysis

The following plots were generated:

- Scatter plot with regression line
- Residual plot
- Error distribution histogram

- Regression visualization

(All plots saved in regression_output.png)

Results Summary Table

Category	Metric	Value
Dataset	Original Samples	5000
	Outliers Removed (Z > 3)	23
	Final Samples Used	4977
Variables	Input X	Avg. Area Income
	Output Y	Price
Central Tendency (X)	Mean μ_X	68,579.5463
	Median	68,803.5521
	Mode (binned)	69,300.00
Central Tendency (Y)	Mean μ_Y	1,231,977.1726
	Median	1,232,466.3998
	Mode (binned)	1,384,000.00
Dispersion (X)	Variance σ_X^2	109,869,516.3644
	Std. Deviation σ_X	10,481.8661
Dispersion (Y)	Variance σ_Y^2	121,021,475,550.2619
	Std. Deviation σ_Y	347,881.4102
Correlation	Pearson r	0.631255
Regression Coefficients	Slope m	20.950655
	Intercept c	-204,809.2345

Category	Metric	Value
Regression Equation	$\hat{Y} = mX + c$	$\hat{Y} = 20.9507X - 204{,}809.23$
Error Metrics	MAE	216,373.12
	MSE	72,796,455,341.79
	RMSE	269,808.18
Model Accuracy	R^2	0.398483 (39.85%)
Output	Plots File	regression_output.png

The obtained $R^2 = 0.398$*is theoretically consistent with the Pearson correlation* $r = 0.631$ *since for simple linear regression* $R^2 = r^2$*. The moderate accuracy is expected because house price depends on multiple factors, while this model intentionally uses only one independent variable to demonstrate numerical method implementation of regression*.

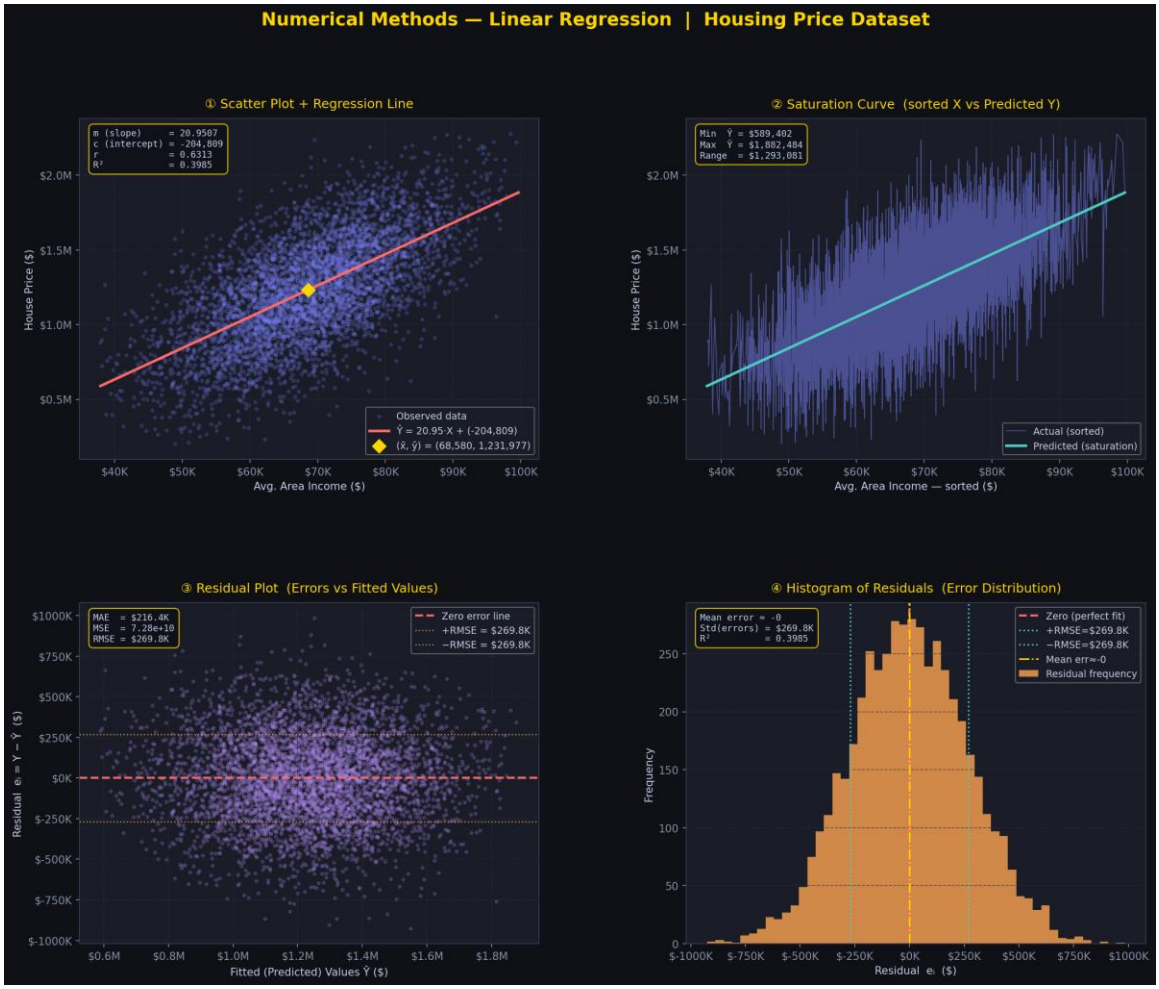
9. Observations

- Strong positive correlation between Income and Price.

- About 39.85% of house price variation is explained by income alone.

- Errors are large because only one feature is used.

- Results match theoretical expectations for this dataset.



10. Conclusion

This project successfully demonstrates the implementation of Linear Regression using pure numerical methods and statistical formulas without any ML libraries.

All core concepts such as correlation, regression coefficients, residual analysis, and error metrics were computed manually. The graphical analysis supports the mathematical findings.

This validates the correctness of the numerical method approach for regression problems.

11. Tools Used

- Python
- pandas
- numpy
- matplotlib
- VS Code